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Unraveling Nitrate Source Dynamics in Megacity Rivers Using an Integrated Machine Learning–Bayesian Isotope Framework

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Abstract

Rapid urbanization has intensified nitrate pollution in megacity rivers, posing severe challenges to urban water governance and sustainable nitrate management. This study presents nitrate dual-isotope signatures ($\delta^{15}N\text{-}NO_3^-$ and $\delta^{18}O\text{-}NO_3^-$) from surface water samples collected during the wet season from the Yongding River (YDR) and Chaobai River (CBR) in the Beijing–Tianjin–Hebei megacity region of North China. Average concentrations of nitrate (as NO_3^-) were 8.5 mg/L in YDR and 12.7 mg/L in CBR. The $\delta^{15}N\text{-}NO_3^-$ and $\delta^{18}O\text{-}NO_3^-$ values varied from 6.1‰ to 19.1‰ and −1.1‰ to 10.6‰, respectively. The spatial distribution of NO_3^-/Cl^- ratios and isotopic data indicated mixed sources, primarily sewage and manure in downstream sections and agricultural inputs in upstream areas. Isotopic evidence revealed widespread nitrification processes and could have potentially localized denitrification under low-oxygen conditions in the lower YDR. Bayesian mixing model (MixSIAR) results indicated that sewage and manure constituted the main nitrate sources (49.4%), followed by soil nitrogen (23.7%), chemical fertilizers (19.2%), and atmospheric deposition from rainfall (7.7%). The self-organizing map (SOM) further revealed three nitrate regimes, including natural and agricultural, mixed, and sewage dominated conditions, indicating a clear downstream gradient of increasing anthropogenic influence. The results suggest that efficient nitrogen management in megacity rivers requires improving biological nutrient removal in wastewater treatment, regulating fertilizer application in upstream areas, and maintaining ecological base flow for natural denitrification. This integrated framework provides a quantitative basis for nitrate control and supports sustainable water governance in highly urbanized watersheds.

Keywords: megacity rivers; nitrate source appointment; dual nitrate isotopes; self-organization map; Bayesian mixing model; Beijing–Tianjin–Hebei region

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Academic Editor: Daniel D. Snow

Received: 4 December 2025
Revised: 26 December 2025
Accepted: 29 December 2025
Published: 1 January 2026

1. Introduction

Urbanization has profoundly reshaped the hydrological and biogeochemical cycles of river systems, leading to intensified nutrient loading and water quality degradation in megacities worldwide [1,2]. Among various pollutants, nitrate (NO_3^-) has become a critical concern due to its high mobility, ecological risks, and potential health impacts, particularly because it can be reduced to nitrite under anaerobic conditions in the human body, which is directly responsible for disorders such as methemoglobinemia and thyroid dysfunction [3–6]. The excessive accumulation of nitrate in urban rivers can deteriorate aquatic ecosystem health and contribute to eutrophication and greenhouse gas emissions [2], posing serious challenges to riverine ecological sustainability [7]. Therefore, understanding the sources and transformation processes of nitrogen is essential for maintaining riverine ecological stability and promoting the sustainable development of urban water systems [8].

Megacity river systems, characterized by high population density, fragmented land use, and complex pollution sources, represent hotspots of nitrogen enrichment [9,10]. The overlapping influence of domestic wastewater, industrial effluents, and agricultural runoff makes it difficult to trace nitrate sources and to understand the mechanisms governing nitrogen cycling [11,12]. Although numerous studies have quantified nitrate levels in major Chinese rivers such as the Weihe, Haihe, and Pearl Rivers [13–15], systematic investigations that integrate multi-source identification with transformation processes in megacity river networks remain limited. This knowledge gap hinders the formulation of effective strategies for integrated nitrogen management in rapidly urbanizing regions. Therefore, nitrate dynamics in megacity rivers require more systematic and long-term investigation [16].

Currently, hydrochemical analysis coupled with nitrate dual isotopes ($\delta^{15}N$-NO_3^- and $\delta^{18}O$-NO_3^-) has been proven to be an effective way for investigating nitrate origins and transformation mechanisms in fluvial systems [17,18], and has been extensively utilized in numerous Chinese watersheds [19–21]. Hydrochemical analyses infer the behavior and provenance of nitrate through evaluating the concentrations and ionic ratios of key ions (e.g., NO_3^-/Cl^-, NO_3^-/Na^+), which serve as indicators of nitrate dynamics and sources [16,22,23]. Meanwhile, isotope-based techniques utilize the distinct $\delta^{15}N$ and $\delta^{18}O$ signatures of potential sources to distinguish among them and to reveal in situ biogeochemical transformations [20,21,24]. Microbial activities such as nitrification and denitrification induce systematic isotopic fractionation, resulting in characteristic variations in the $\delta^{15}N$ and $\delta^{18}O$ values of the remaining nitrate pool [25]. Previous studies have shown that different nitrate sources exhibit distinct isotopic signatures, and $\delta^{15}N$ and $\delta^{18}O$ values vary by source: inorganic fertilizers (−6‰ to +6‰; +17‰ to +25‰), manure (5‰ to 25‰; 5‰ to 7‰), sewage effluent (4‰ to 19‰; −5‰ to +10‰), atmospheric nitrate (−13‰ to +13‰; +25‰ to +75‰), and soil nitrogen (0‰ to 8‰; −10‰ to +10‰) [26–29]. However, these conventional methods often struggle with source quantification due to overlapping isotopic signals, especially in megacity rivers. To address these challenges, a combination of pattern recognition and probabilistic modeling was adopted in this study. The Self-Organizing Map (SOM), an unsupervised neural network, was used to identify spatial and compositional patterns in hydrochemical and isotopic data, thereby distinguishing river segments with similar nitrate characteristics and mitigating the influence of signal overlap [30,31]. The Bayesian mixing model (MixSIAR), based on a hierarchical Bayesian framework, was subsequently applied to quantify source contributions by linking observed isotopic values with end-member signatures while explicitly accounting for variability, uncertainty, and prior information [32–34]. Although this model has demonstrated promising performance in agricultural and small urban catchments where source signals are relatively distinct [24], integrating SOM provides additional spatial structure and improves interpretability under the highly heterogeneous conditions of megacity rivers. This combined SOM-MixSIAR framework therefore offers a more robust approach for identifying nitrate regimes and quantifying source contributions in complex urban watersheds [32,35].

The Yongding River (YDR) and Chaobai River (CBR) are two major rivers traversing the Beijing–Tianjin–Hebei megacity cluster, providing essential water resources for urban supply, agriculture, and ecological restoration [36]. Rapid urbanization, industrialization, and intensive land-use conditions have sharply increased water utilization and pollutant inputs, leading to persistent water scarcity and substantial environmental pressure in this region [37]. Despite increasing concern about nitrate pollution in these rivers, most studies were focused on individual tributaries or sections of major rivers such as the Chao River and Jian River [38,39], with relatively few efforts examining nitrate distribution and sources across entire river systems. Therefore, it is crucial to understand the nitrate sources and transformation mechanisms in these rivers systematically, which could benefit nitrogen pollution control and promote urban water sustainability in megacities facing water stress. This study systematically analyzed nitrate distribution patterns and controlling factors in YDR and CBR via an integrated framework combined with hydrochemistry, dual isotopes, and a Bayesian mixing model, and aims to (1) characterize the spatial variation in nitrate concentrations and isotopic compositions; (2) identify and quantify the primary nitrate sources; (3) assess the influence of nitrogen transformation processes on nitrate dynamics. These findings provide scientific support for managing nitrate pollution in urban rivers of the Beijing–Tianjin–Hebei megacity cluster and for promoting integrated water quality improvement and nitrogen management in rapidly urbanizing regions.

2. Materials and Methods

2.1. Study Area

The YDR and CBR basins are located in the Beijing–Tianjin–Hebei megacity clusters of North China, with a total drainage area of approximately 35,500 km^2 (Figure 1). Both rivers originate from mountainous areas upstream of major reservoirs and flow southeast into the Bohai Sea. The YDR is regulated by the Guanting Reservoir, and the CBR by the Miyun Reservoir, both of which serve as crucial drinking water sources for Beijing and play key roles in regional water supply and flow regulation. The region has a typical monsoon climate, receiving 400–600 mm of annual precipitation, with over 80% occurring between June and September [36]. Land use across the basins presents a distinct spatial gradient. Woodland, grassland, and dryland agriculture dominate the upstream areas, whereas midstream and downstream areas transition from a mix of cultivated land and suburban settlements to dense urban infrastructure dominated by construction land. Due to the rapid developments of population and economic, this megacity area has become one of the most urbanized areas in China [40]. The region features a diverse industrial structure, with Beijing focusing on high-tech and service sectors, while Tianjin hosts large-scale manufacturing and port-related industries [41]. The YDR and CBR play essential roles in regional water supply, but both rivers face ongoing hydrological and ecological pressures [42]. Management measures (i.e., ecological water replenishment projects) have partially restored flow regimes, especially in the YDR [43], yet rising urban and industrial activities continue to introduce substantial pollutant loads, posing ongoing challenges to water quality management in the basins.

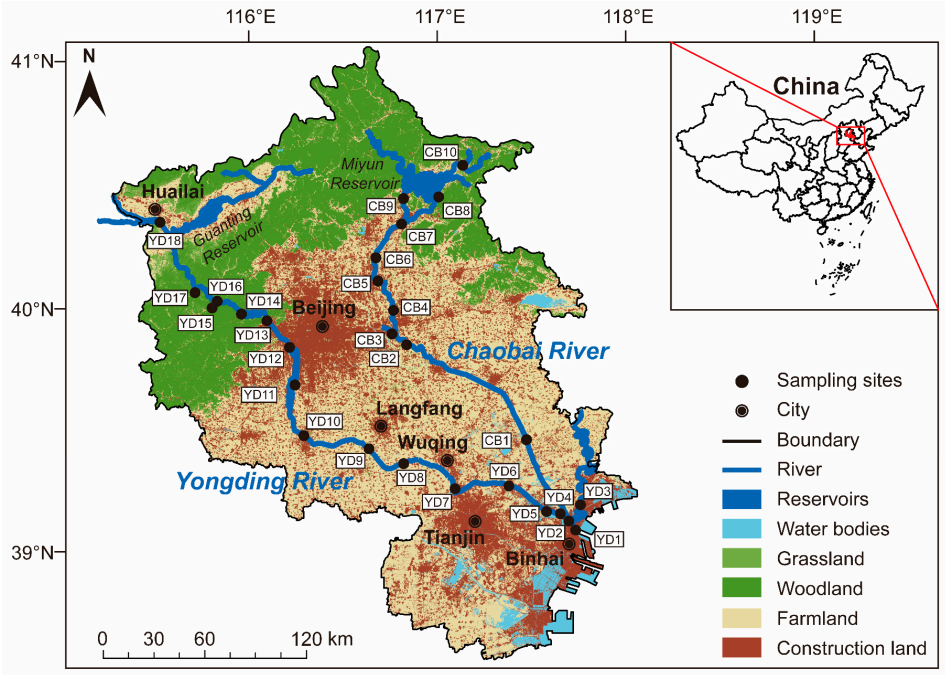

Figure 1. Map showing the location, land-use, and sampling sites of the two rivers.

2.2. Sampling and Analysis

Locations of the study area and sampling sites are illustrated in Figure 1. The study area was located in the center of the Beijing–Tianjin–Hebei region, which encompasses the section of YDR from the confluence of the Yanghe River and Sanggan River to the estuary and the section of CBR from the Miyun Reservoir to the confluence with the YDR in Binhai. According to the variations in land-use and anthropogenic activities, 18 sampling sites were established along the YDR and 10 along the CBR. These sites were further classified into three segments: downstream (YD1~YD7; CB1~CB2), midstream (YD8~YD13; CB3~CB6), and upstream (YD14~YD18; CB7~CB10), respectively. The sampling campaign including collection of river water samples and in-field determination was carried out during July 2021. All water samples were collected from the central flow at approximately 0.3~0.5 m below the water surface to represent well-mixed surface water conditions. After collection, the sample was immediately filtered through a 0.22 µm mixed-fiber membrane using a glass vacuum filtration unit. Filtrates were transferred into high-density polyethylene (HDPE) bottles, and rinsed three times with the sample water prior to use. Specifically, subsamples for cation analysis were acidified with ultrapure HCl to maintain pH < 2. All the containers and membranes mentioned above are pre-cleaned by HCl and MilliQ-water (Merck Millipore, Burlington, MA, USA) in the lab before the campaign, and all prepared samples were stored at 4 °C away from light until subsequent laboratory measurements.

Water physicochemical parameters and major ions of the YDR and CBR have been reported in our previous studies [36,44]. In brief, water physicochemical parameters were measured by a YSI ProPlus portable multiparameter instrument (YSI, a Xylem brand, Yellow Springs, OH, USA). Cations and anions were determined by inductively coupled plasma optical emission spectrometry (ICP-OES, Thermo Fisher Scientific, Waltham, MA, USA) and ion chromatograph (ICS-600, Thermo Fisher Scientific, Waltham, MA, USA), respectively. The Nessler reagent method was employed for the measurement of NH_4^+ concentration. Replicate samples and certified reference materials were used to verify measurement accuracy, maintaining deviations within ±5%.

Isotopic compositions of nitrate ($\delta^{15}N\text{-}NO_3^-$ and $\delta^{18}O\text{-}NO_3^-$) were determined using the denitrifier method. According to the NO_3^- concentration in each sample, cultured *Pseu-*

domonas chlororaphis subsp. aureofaciens (ATCC 13985, American Type Culture Collection, Manassas, VA, USA) were employed to reduce NO_3^- to N_2O [45,46]. Prior to the reaction, high-purity N_2 gas was passed through the vials for approximately three hours to maintain an oxygen-free condition. To suppress microbial activity and eliminate CO_2, 0.2 mL of 10 mol/L NaOH was added to each sample [46]. The resulting N_2O was purified using a Trace Gas Pre-concentrator and then analyzed on IRMS. Four international standards were employed for the calibration, which including USGS-32, USGS-34, USGS-35, and IAEA-N3, Yielding analytical precisions of $\delta^{15}N\text{-}NO_3^-$ and $\delta^{18}O\text{-}NO_3^-$ were within ±0.3‰ and ±0.5‰, respectively. The isotopic composition was expressed as [47]:

$$\delta\ (‰) = [(R_{sample}/R_{standard}) - 1] \times 1000, \tag{1}$$

Here, R_{sample} and $R_{standard}$ denote the $^{15}N/^{14}N$ or $^{18}O/^{16}O$ isotope ratios measured in the samples and reference materials. The values of $\delta^{15}N$ and $\delta^{18}O$ are relative to atmospheric N_2 and Vienna Standard Mean Ocean Water (VSMOW).

2.3. Nitrate Source Identification Framework

2.3.1. Bayesian Isotope Mixing Model

The MixSIAR model applies a Bayesian inference framework to quantify the relative inputs of multiple potential sources in isotope-based mixing systems. Rather than relying on deterministic calculations, it estimates source proportions through probabilistic simulations that explicitly integrate measurement errors, natural variability, and uncertainties in both source signatures and observed isotope compositions [48]. Generally, the model can be described as the following equations [49]:

$$X_{ij} = \Sigma_{k=1}^{k} P_k \left(S_{jk} + c_{jk}\right), \tag{2}$$

$$S_{jk} \sim N\ (\mu_{jk}, \omega_{jk}{}^2), \tag{3}$$

$$c_{jk} \sim N\ (\lambda_{jk}, \tau_{jk}{}^2), \tag{4}$$

$$\varepsilon_{ij} \sim N\ (0, \sigma_j{}^2), \tag{5}$$

where X_{ij} is the observed stable isotope ratio of the sample; k is the number of potential sources; P_k is the proportional contribution of source k to the river water; S_{jk} is the true stable isotope value of source k in individual j, μ_{jk} is the mean isotope value, $\omega_{jk}{}^2$ is the variance representing the natural variability of the source; c_{jk} is the isotopic fractionation correction term, λ_{jk} is the mean correction factor, $\tau_{jk}{}^2$ is the variance of the correction factor representing uncertainty in the adjustments; ε_{ij} represents the independent error component that accounts for measurement noise and observational variability, and is treated as a normally distributed term with a mean of 0 and a variance of $\sigma_j{}^2$.

2.3.2. Self-Organization Map

By preserving the inherent topological structure of the dataset, the SOM arranges samples on a two-dimensional lattice. This unsupervised neural network accomplishes this by mapping patterns from a high-dimensional feature space onto a simplified grid representation [50]. This method has been widely used for pattern recognition and clustering in environmental datasets where complex nonlinear interactions exist among variables [51].

In this study, a dataset containing major ions (Na^+, K^+, Mg^{2+}, Ca^{2+}, Cl^-, SO_4^{2-}, HCO_3^-, NO_3^-), physicochemical parameters (pH, DO, TDS), and dual-isotope compositions ($\delta^{15}N\text{-}NO_3^-$ and $\delta^{18}O\text{-}NO_3^-$) was normalized using z-score standardization prior to SOM analysis. The SOM was trained using the kohonen package in R 4.3.0, with a hexagonal grid structure to preserve neighborhood relationships. The optimal grid size was determined through multiple trials to minimize quantization and topographic errors, and the learning rate was gradually decreased from 0.05 to 0.01 over 500~1000 iterations to ensure convergence [52].

After training, each sample was assigned to its best-matching neuron on the SOM grid. The resulting codebook vectors were subsequently analyzed using k-means clustering to group neurons with similar properties. This combined SOM-k-means framework enabled data-driven pattern recognition and was used to provide the structural basis for subsequent isotope-based source apportionment in the MixSIAR model.

2.4. Data Analysis

A schematic framework is presented in Figure 2 to illustrate the overall design, data processing, and analytical workflow of this study. All statistical analyses were conducted using SPSS 26.0 and OriginPro 2021. To evaluate differences in nitrate concentrations between the YDR and CBR, independent sample *t*-tests were performed. To assess spatial variability among the upper, middle, and lower reaches within each river, one-way ANOVA was applied, followed by post hoc comparisons. In addition, linear regression analyses were performed to examine the relationships between nitrate isotopic compositions and ion ratios, to infer the occurrence and strength of denitrification processes. A significance level of $p < 0.05$ was used for all statistical tests.

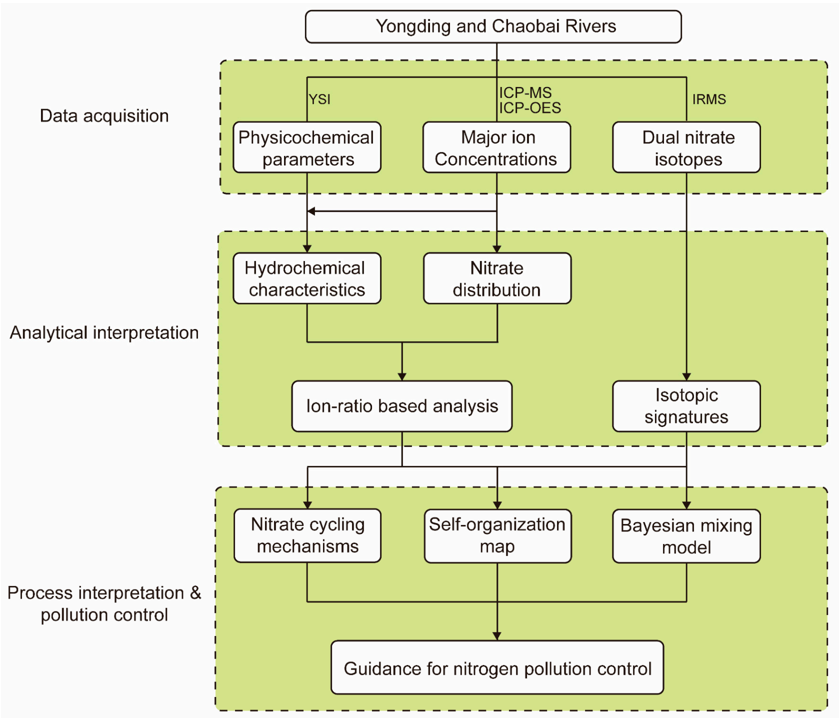

Figure 2. Methodological framework of the study.

3. Results and Discussion

3.1. Physicochemical Characteristics of River Water

The physicochemical characteristics provide the hydrochemical basis for interpreting nitrate behavior and anthropogenic influence in the studied megacity rivers. The statistical characteristics of the water physicochemical parameters of the YDR and CBR are listed in Table 1, and the detailed data are listed in Tables S1 and S2. Water temperature ranged from 18.4 °C to 31.6 °C in the YDR and from 25.6 °C to 31.5 °C in the CBR, with mean values of 26.8 °C and 29.1 °C, respectively ($p > 0.05$). The pH value ranged from 7.6 to 9.0 in YDR and from 7.7 to 9.2 in CBR, with average values of 8.2 and 8.4, respectively. Although both rivers exhibited alkaline conditions, pH values of the YDR showed a significant decreasing trend from the upstream mountainous reaches to the downstream urban sections ($p < 0.05$), whereas in the CBR was relatively uniform ($p > 0.05$). Dissolved oxygen (DO) is deeply related to the biogeochemical processes in aquatic systems and acts as an essential factor for water chemistry [53,54]. The average DO concentrations in the YDR and CBR were 5.8 mg/L and 8.2 mg/L, with corresponding DO saturation levels of 73.9% and 104.4%, respectively. One-way ANOVA showed significant spatial differences in DO concentrations and saturation among YDR segments ($p < 0.05$), with the lowest values observed in the downstream reaches (YD1~YD7). This downstream decline in DO indicates intensified organic matter decomposition and oxygen depletion driven by excessive nutrient inputs in urbanized areas, reflecting strong anthropogenic disturbance [55]. Supporting this observation, the YDR exhibited a higher TDS concentration (571.2 mg/L) compared to the CBR (355.4 mg/L) ($p < 0.05$), exceeding the global riverine average (283 mg/L). The elevated TDS, accompanied by low DO, clearly indicates substantial anthropogenic solute inputs and limited dilution capacity in the lower YDR, where increased organic and nutrient loading enhances oxygen consumption.

The concentrations and spatial distribution characteristics of major ions in the two rivers have been elucidated in our previous studies [36,44]. The dominant water chemical types of the YDR and CBR were $SO_4^{2-}\cdot Cl^-$-Na^+ and HCO_3^--$Ca^{2+}\cdot Mg^{2+}$, respectively. Despite anthropogenic influence, weathering-derived ions (e.g., Ca^{2+}, Mg^{2+}, and K^+) showed stable distributions, suggesting that natural lithological weathering remains a key background contributor [48]. In contrast, anthropogenically related ions (Cl^- and SO_4^{2-}) exhibited significant downstream increases in both rivers ($p < 0.05$), consistent with intensified urban and industrial activities in lower reaches. These solutes are typically linked to domestic sewage, industrial effluents, and atmospheric inputs [56], confirming their anthropogenic origin. These patterns support the use of ion indicators in identifying pollutant inputs and interpreting nitrate dynamics [57].

Table 1. Statistical characteristics of the physicochemical parameters and isotope composition of the water samples in YDR and CBR.

Items	YDR				CBR			
	Min	Max	Mean	SD	Min	Max	Mean	SD
T (°C)	18.4	31.6	26.8	3.0	25.6	31.5	29.1	2.0
pH	7.6	9.0	8.2	0.5	7.7	9.2	8.4	0.5
DO (mg/L)	1.8	11.1	5.8	2.9	5.9	12.8	8.2	1.8
DO (%)	22.6	125.0	73.9	35.4	72.0	170.1	104.4	27.2
NO_3^- (mg/L)	1.8	23.2	8.5	6.0	3.3	29.5	12.7	8.3
NH_4^+ (mg/L)	0.1	0.9	0.3	0.3	0.1	0.8	0.4	0.3
$\delta^{15}N$-NO_3^- (‰)	6.1	19.1	10.4	3.7	6.3	16.8	10.4	3.5
$\delta^{18}O$-NO_3^- (‰)	−1.1	8.1	5.0	2.8	5.9	10.6	7.9	1.6

3.2. Spatial Variations of Nitrogen and Nitrate Dual Isotopes

The spatial distribution of dissolved inorganic nitrogen and nitrate isotopes reveals strong anthropogenic influence and hydrological modulation across the YDR and CBR. The statistical characteristics of NH_4^+, NO_3^-, and isotopic signatures of the YDR and CBR are listed in Table 1. Concentrations of NH_4^+ were varied 0.1~0.9 mg/L in YDR and 0.1~0.8 mg/L in CBR, averaged 0.3 and 0.4 mg/L, respectively. Both rivers had low NH_4^+ levels, and displayed no statistical significance ($p > 0.05$). However, NO_3^- concentrations in both rivers displayed higher levels, from 1.8 to 23.2 mg/L (averaged 8.5 mg/L) in YDR and from 3.3 to 29.5 mg/L (averaged 9.4 mg/L) in CBR. Isotopic composition of the YDR ranged from 6.1‰ to 19.1‰ for $\delta^{15}N-NO_3^-$ (averaged 10.4‰), and from −1.1‰ to 8.1‰ for $\delta^{18}O-NO_3^-$ (averaged 5.0‰), while in CBR ranged from 6.3‰ to 16.8‰ and from 5.9‰ to 10.6‰, averaged 10.4‰ and 7.9‰, respectively. To compare the relative importance of oxidized and reduced inorganic nitrogen species, dissolved inorganic nitrogen (DIN) was expressed as the sum of NO_3^- and NH_4^+ concentrations, as nitrite concentrations were negligible in all samples. The proportion of NO_3^- in DIN ranged from 75% to 99% in the two rivers, with an average of 96%, suggesting that NO_3^- is the dominant form of DIN in both rivers. These values were higher than those in typical agriculturally or naturally influenced rivers such as the Jiulong River (5.8 mg/L), and significantly higher than other rivers worldwide including Mun River (0.8 mg/L) in Thailand, Seine River (2~6 mg/L) in France and Ebro River (9.4 mg/L) in Spain, reflecting enhanced nitrate enrichment under anthropogenic pressure [18,58–60].

Spatial variations in parameters (DO), ions (NO_3^- and Cl^-), and isotopes are displayed in Figure 3. Overall, nitrate concentrations exhibited distinct longitudinal patterns between the two rivers. The nitrate levels increased progressively downstream, particularly through the urbanized zones of Tianjin and Binhai (averaged 9.4 mg/L from YD1 to YD5), peaking at 10.6 mg/L at YD1, which was significantly higher than upstream reaches ($p < 0.05$). This trend corresponded with $\delta^{15}N$ values (averaged 10.2‰) and rising Cl^- concentrations (averaged 207.6 mg/L from YD1 to YD5), reflecting intensified anthropogenic influence [61]. In contrast, the CBR showed irregular or "zigzag" distribution patterns without a significant longitudinal trend ($p > 0.05$), indicating stronger local heterogeneity. This zigzag pattern reflects the combined effects of reservoir retention, tributary inputs, and heterogeneous land use, rather than a simple longitudinal accumulation process. The large spacing between CB1 and CB2, together with reservoir regulation and tributary inflows, likely contributes to the observed spatial discontinuity in nitrate patterns. The highest and lowest NO_3^- concentrations were observed at CB10 and CB9 (13.7 mg/L and 3.3 mg/L, respectively), with a significant difference between the two sites ($p < 0.05$), indicating strong spatial heterogeneity in nitrate levels. Both sites exhibited relatively low Cl^- concentrations (84.7 mg/L at CB10 and 12.8 mg/L at CB9), indicating that nitrate variation was not primarily driven by anthropogenic point-source pollution [58]. CB10 is located upstream of the Miyun Reservoir in an agricultural zone, likely received nitrate from fertilizer application and soil nitrogen mineralization [62], whereas CB9, immediately below the reservoir outlet, showed strong dilution and retention effects [63]. Moreover, CB3 displayed elevated NO_3^- (21.4 mg/L) and $\delta^{15}N-NO_3^-$ (16.8‰) values, attributed to inflow from the Yunchaojian River draining central Beijing, where domestic sewage is the main source [64].

In summary, downstream enrichment of NO_3^- and $\delta^{15}N-NO_3^-$ in the YDR reflects enhanced sewage and industrial inputs, while the more variable pattern in the CBR highlights the regulating role of the Miyun Reservoir and mixed land use. These results are consistent with the spatial distribution of DO and Cl^-, collectively suggesting that nitrate behavior in the two river systems is largely controlled by anthropogenic activities superimposed on hydrological and geomorphological settings. Therefore, quantitative source apportionment using isotopic and Bayesian modeling is essential to further distinguish multiple overlapping nitrate sources, as discussed in the following section.

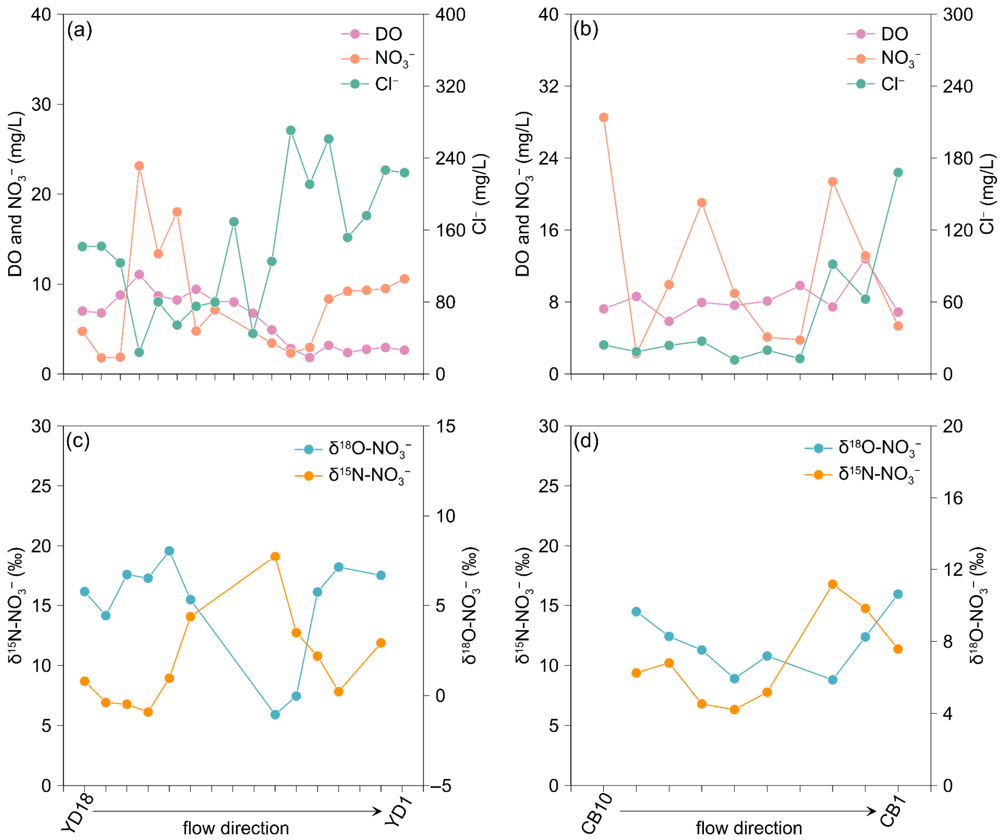

Figure 3. Spatial distribution of ions and isotopic signatures in the two rivers; (**a**) DO, NO_3^-, and Cl^- in YDR; (**b**) DO, NO_3^-, and Cl^- in CBR; (**c**) $\delta^{15}N$-NO_3^- and $\delta^{18}O$-NO_3^- in YDR; (**d**) $\delta^{15}N$-NO_3^- and $\delta^{18}O$-NO_3^- in CBR.

3.3. Nitrate Sources Constrained by Ion-Ratio and Dual Isotopes

Ion-ratio analysis and dual-isotope evidence were jointly applied to distinguish natural and anthropogenic contributions across the two river systems. Currently, hydrochemical analysis has been widely applied in studies on nitrate source apportionment in aquatic systems [65]. Previous studies have improved that the NO_3^-/Cl^- ratio can reflect the type and proportion of nitrogen inputs and transformation [66]. A high NO_3^-/Cl^- ratio (>0.3) generally suggests non-point sources such as agricultural fertilization or soil nitrogen mineralization, whereas a low ratio (<0.1) may be associated with domestic sewage inputs, Cl-rich background water, or nitrate removal processes (e.g., denitrification), because nitrate is a non-conservative species that can be affected by biological uptake and anaerobic microenvironments [32].

The NO_3^-/Cl^- ratios for the YDR and CBR are shown in Figure 4a. In the CBR, most samples exhibited moderate Cl^- concentrations (200~2000 μmol/L) and NO_3^-/Cl^- ratios of 0.1~1.0, implying active nitrogen inputs from agriculture and soil N mineralization. However, site CB1 exhibited a notably high Cl^- concentration and a low NO_3^-/Cl^- ratio, which deviated significantly from the general CBR pattern. This site is located in an area with riverbank aquaculture, where water diverted into enclosed ponds undergoes nitrate removal through algal uptake or denitrification during retention, while Cl^- returns to the river largely unchanged [67,68]. In contrast, YDR samples exhibited overall higher Cl^- concentrations (1000~8000 μmol/L, $p < 0.05$) and lower NO_3^-/Cl^- ratios, consistent with continuous sewage discharge, industrial effluents, or contaminated groundwater inputs. A notable exception was YD15, which exhibited a lower Cl^- concentration and a relatively higher NO_3^-/Cl^- ratio ($p < 0.05$ when compared to downstream YDR sites). This site represents the outlet of Zhaitang Reservoir, and the land-use was mainly composed of forests, bare land, and a few croplands [69]; therefore, the nitrate inputs of this site might be related to soil N and surrounding agricultural activities. These results collectively show that the NO_3^-/Cl^- ratio effectively differentiates natural from anthropogenic signals but cannot fully separate overlapping sources under mixed inputs or hydrological dilution. Therefore, isotopic techniques were used for further verification.

Nitrate dual isotopes provide stable, source-specific signatures, allowing differentiation among sewage, manure, fertilizers, atmospheric deposition, and soil N [26,70]. Meanwhile, isotopic signatures can provide strong support in identifying biogeochemical transformation processes, such as denitrification and nitrification, making them become one of the most reliable tools currently available for nitrate source apportionment in urban aquatic systems [71]. Figure 4b presents the $\delta^{15}N\text{-}NO_3^-$ and $\delta^{18}O\text{-}NO_3^-$ values for the YDR and CBR, along with the ranges for different nitrate sources reported in previous studies [26,70]. In the CBR, nitrate was mainly influenced by a combination of agricultural runoff and localized anthropogenic discharges, as reflected by isotope values positioned near the boundary between the soil nitrogen and manure/sewage end-members. In particular, CB1 ($\delta^{15}N\text{-}NO_3^-$ = 13.8‰, $\delta^{18}O\text{-}NO_3^-$ = 11.6‰) showed strong enrichment, consistent with nitrogen from aquaculture effluents, whereas upstream sites indicated more mixed agricultural and natural influences [67]. Nitrate in the YDR exhibited pronounced spatial variability. Upstream sites (from YD12 to YD15) had low $\delta^{15}N$ (<8‰), typical of natural soil N and fertilizer-derived nitrate, while midstream to downstream sites (from YD5 to YD12) showed $\delta^{15}N$ values up to 19‰, suggesting increasing sewage and manure contributions [72]. Statistical comparison indicated that $\delta^{15}N\text{-}NO_3^-$ averages indicated no significant difference between the two rivers ($p > 0.05$), while $\delta^{18}O\text{-}NO_3^-$ values were significantly higher in the CBR ($p < 0.05$), implying stronger atmospheric influence and in situ nitrification processes. Several samples in both rivers were observed along or between the 1:1 and 2:1 enrichment slopes in the dual-isotope space, indicating possible isotopic fractionation during nitrate reduction [25]. Although this section focuses on source attribution, these patterns suggest that biogeochemical processes such as denitrification may be simultaneously occurring and influencing isotopic signals. The following section will provide a detailed explanation of potential nitrogen transformation processes in the two rivers.

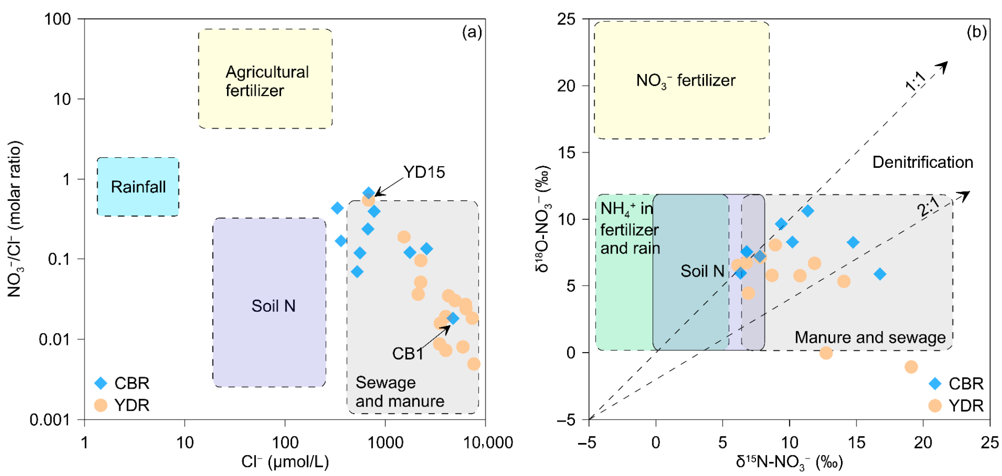

Figure 4. Identification of nitrate sources based on ion ratios and dual isotopes; (**a**) NO_3^-/Cl^- versus Cl^-; (**b**) $\delta^{15}N\text{-}NO_3^-$ versus $\delta^{18}O\text{-}NO_3^-$.

3.4. Nitrate Transformation Processes

To explore nitrate behavior beyond source identification, hydrochemical and isotopic indicators were used to evaluate in-stream nitrification and denitrification processes in the

YDR and CBR, and the results are shown in Figure 5. Redox conditions inferred from DO and pH serve as fundamental controls on nitrogen transformation pathways [73]. Ideal conditions for nitrification generally correspond to DO > 6 mg/L and pH between 7.5 and 8.5, while denitrification is favored under DO < 2 mg/L and pH between 6.5 and 7.5 [74]. In Figure 5a, all samples of CBR fell within or near the optimal nitrification zone, suggesting that aerobic conditions prevail and are suitable for ammonia oxidation. In contrast, several YDR samples, particularly from middle to downstream reaches, were located within or near the low-DO and near-neutral pH region corresponding to the denitrification domain. This spatial distinction suggests a greater potential for denitrification in the YDR compared to the CBR.

During nitrification, the oxygen atoms incorporated into the newly formed nitrate originate from both water (H_2O) and atmospheric O_2. Therefore, the relationship between $\delta^{18}O\text{-}NO_3^-$ and $\delta^{18}O\text{-}H_2O$ provides an effective means of identifying whether nitrate is produced through in situ nitrification. Previously, two principal oxygen incorporation models have been proposed to describe the isotopic composition of nitrified nitrate. The first is a simplified model in which 2/3 of the oxygen in NO_3^- is derived from H_2O and 1/3 from O_2 [26,70], and the second is a more detailed two-step model, which considers the sequential oxidation of NH_4^+ to NO_2^- and then to NO_3^-, resulting in a ratio of 5/6 H_2O and 1/6 O_2 for the oxygen incorporated into NO_3^- [75]. These two models define a theoretical $\delta^{18}O\text{-}NO_3^-$ range that can be used to evaluate whether nitrification is a dominant nitrate source. The calculations can be described as follows:

$$\delta^{18}O\text{-}NO_3^- = 2/3\ \delta^{18}O\text{-}H_2O + 1/3\ \delta^{18}O_{air}, \quad (6)$$

$$\delta^{18}O\text{-}NO_3^- = 5/6\ \delta^{18}O\text{-}H_2O + 1/6\ \delta^{18}O_{air}, \quad (7)$$

The $\delta^{18}O\text{-}H_2O$ of the YDR and CBR were measured from the same set of samples and have been reported by previous studies [76], and the results are shown in Figure 5b. All samples from both the YDR and CBR were plotted above the two theoretical nitrification lines, indicating that nitrification is the dominant process contributing to nitrate formation across the area. The $\delta^{18}O\text{-}NO_3^-$ generally followed the trend expected from oxygen incorporation during nitrification, with most samples reflecting contributions from both water and DO. This interpretation is further supported by the generally low NH_4^+ concentrations observed in both rivers, suggesting that reduced nitrogen was efficiently oxidized to nitrate under predominantly oxic conditions. Nitrite concentrations were consistently low and therefore not explicitly discussed, indicating that NO_2^- acted mainly as a transient intermediate during nitrification rather than accumulating in the water column. Notably, samples YD6 and YD7 fall under the first oxygen incorporation models and exhibit markedly low DO concentrations (~2 mg/L), indicating that nitrification may be limited at these sites.

Denitrification reduces nitrate in aquatic environments while simultaneously enriching the remaining NO_3^- in heavier isotopes due to the preferential removal of lighter isotopes by microbial processes [77]. Accordingly, the relationships between nitrate isotopes and $\ln(NO_3^-/Cl^-)$ can be used to assess the occurrence and intensity of denitrification [78], and are displayed in Figure 5c and Figure 5d, respectively. In Figure 5c, the $\delta^{15}N\text{-}NO_3^-$ in YDR exhibited a significant negative correlation with $\ln(NO_3^-/Cl^-)$ ($R^2 = 0.47$, $p < 0.05$), indicating isotopic enrichment of residual nitrate due to microbial denitrification. The calculated enrichment factor ($\varepsilon \approx -1.9‰$) fell within the expected range for aquatic systems undergoing open-system Rayleigh-type fractionation. In contrast, no significant correlation was observed in the CBR ($R^2 = 0.15$), suggesting that $\delta^{15}N\text{-}NO_3^-$ variation was dominated by multiple overlapping nitrate sources rather than transformation. Notably, the middle and downstream sections of the YDR showed an even stronger relationship

(R^2 = 0.81, p < 0.05), suggesting more active denitrification processes along the downstream gradient. In contrast, no significant correlation was observed in the CBR (R^2 = 0.15), implying that $\delta^{15}N\text{-}NO_3^-$ variability was primarily influenced by source mixing rather than in situ transformation.

In Figure 5d, an opposite pattern was observed for $\delta^{18}O\text{-}NO_3^-$. In the CBR, a strong negative correlation with $\ln(NO_3^-/Cl^-)$ was detected (R^2 = 0.57, p < 0.05), while the YDR showed a weaker and statistically insignificant relationship (R^2 = 0.23, p = 0.13), indicating that $\delta^{18}O\text{-}NO_3^-$ was not substantially enriched in response to nitrate depletion. However, when only the middle and downstream YDR sites were considered, $\delta^{18}O\text{-}NO_3^-$ exhibited a remarkably strong positive correlation with $\ln(NO_3^-/Cl^-)$ (R^2 = 0.95, p < 0.01), with a steep slope (~4.76) exceeding the typical 2:1 $\delta^{15}N$:$\delta^{18}O$ enrichment ratio. This suggests that denitrification was accompanied by oxygen isotope exchange between nitrate intermediates and ambient water, leading to altered $\delta^{18}O$ dynamics. Additionally, samples YD6 and YD7, which showed both low DO (~2 mg/L) and extremely low $\delta^{18}O\text{-}NO_3^-$ values, appear to represent localized denitrification hotspots. Both sites are located along low-energy reaches characterized by muddy, organic-rich sediments that promote suboxic to anoxic microzones within the sediment-water interface [57]. These conditions, coupled with potential anthropogenic nitrogen inputs and limited water exchange, create favorable environments for heterotrophic denitrifiers [79]. Under such settings, exchange-driven isotopic dilution may further mask the $\delta^{18}O$ enrichment signature. These findings underscore the spatial heterogeneity of nitrate transformation processes and highlight the critical role of local redox and hydrological conditions in shaping isotopic patterns in urban rivers.

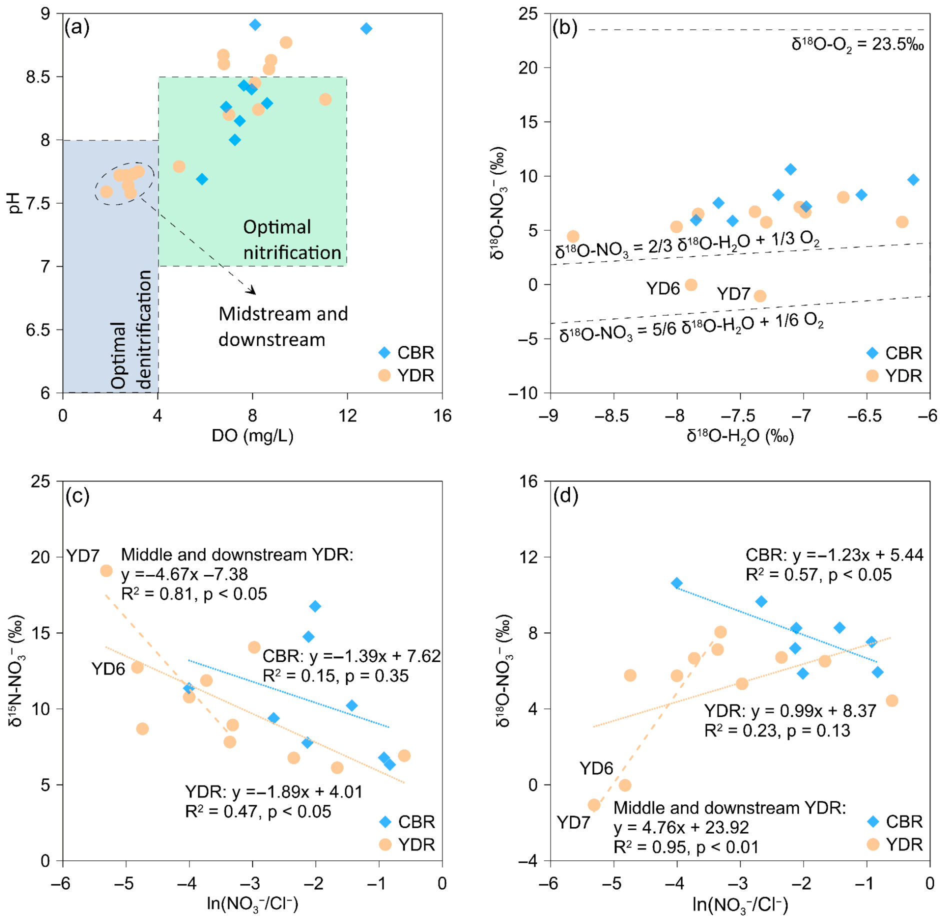

Figure 5. Nitrate transformation processes inferred from hydrochemical and isotopic indicators; (**a**) DO versus pH; (**b**) $\delta^{18}O\text{-}H_2O$ versus $\delta^{18}O\text{-}NO_3^-$; (**c**) $\ln(NO_3^-/Cl^-)$ versus $\delta^{15}N\text{-}NO_3^-$; (**d**) $\ln(NO_3^-/Cl^-)$ versus $\delta^{18}O\text{-}NO_3^-$.

3.5. Source Apportionment Using SOM-MixSIAR

Evidence from ion ratios and dual isotopes has indicated that the multiple nitrate sources altered the nitrate distributions in YDR and CBR. While the main conclusions could be inferred using conventional isotope and hydrochemical analyses, the machine learning-assisted framework improves pattern recognition under complex and overlapping source conditions, particularly in megacity rivers with high spatial heterogeneity. To further quantify source contributions of riverine nitrate, a SOM-guided Bayesian mixing model was applied using four potential end-members: chemical fertilizer, rainwater, sewage and manure, and soil N [80,81]. Isotopic signatures of these end-members were referenced from previous studies focused on Beijing [82–85] and are listed in Table 2. The results are displayed in Figure 6.

Table 2. Isotopic compositions of potential nitrate sources used in the MixSIAR.

Sources	δ^{15}N-NO_3^- (‰)	δ^{18}O-NO_3^- (‰)
	Mean ± Std	Mean ± Std
Rainwater	−6.4 ± 5.5	46.6 ± 22.6
Soil N	1.8 ± 1.6	1.7 ± 0.5
Chemical fertilizer	0.9 ± 2	3 ± 1.4
Sewage and manure	14.3 ± 1.9	6.7 ± 2.5

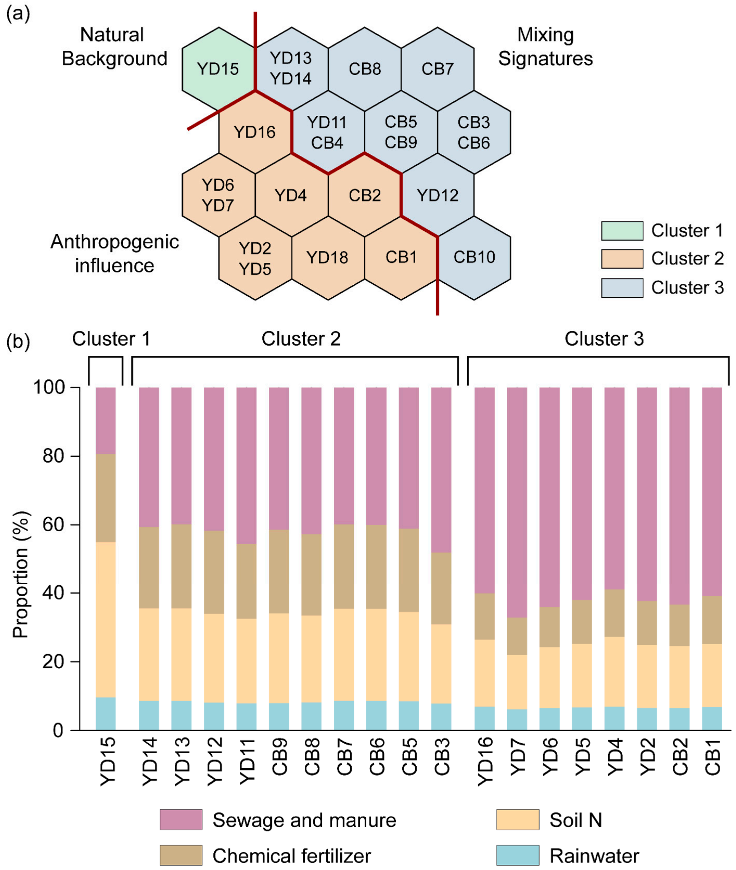

Figure 6. Sources apportionment based on SOM-MixSIAR coupling: (**a**) SOM classification of all samples; (**b**) proportional contributions of nitrate sources via MixSIAR.

The SOM-MixSIAR framework identified clear spatial transitions in nitrate sources across the two rivers. On average, sewage and manure contributed the largest proportion of nitrate (about 49.4% across all sites), followed by soil N (23.7%), chemical fertilizer

(19.2%), and rainwater (7.7%). The dominance of anthropogenic sources (sewage and fertilizer together accounting for over 60%) indicates that nitrate inputs in both rivers were mainly human-derived. The SOM classification provided a process-consistent structure for interpreting these results by separating the sampling sites into three clusters corresponding to natural–agricultural, mixed, and sewage-dominated regimes.

Cluster 1 includes YD15, which represents the outflow from the Zhaitang Reservoir receiving mountainous inflows, is characterized by low NO_3^- and Cl^- concentrations, high DO, and light $\delta^{15}N\text{-}NO_3^-$ values. This reach reflects the influence of mountainous inflows and reservoir release, and thus typifies a natural-source regime with minimal anthropogenic impact. The MixSIAR results show that soil N (45%) and chemical fertilizer (26%) were the dominant sources, whereas sewage and manure contributed only 19%. These results suggest that nitrate mainly originated from soil mineralization and fertilizer nitrification in the forest-farmland mosaic of the upper catchment, consistent with the natural background conditions maintained by mountainous inflows [48].

Cluster 2, including the midstream and peri-urban sites (YD11~YD14 and CB3~CB10), represents mixed contributions by agricultural runoff and urban wastewater. Sewage and manure accounted for 43% of the nitrate, while soil N and fertilizer contributed 26% and 24%, respectively. These areas exhibited moderate $\delta^{15}N\text{-}NO_3^-$ values (8~12‰) and relatively high DO, implying active nitrification with limited denitrification. The combination of urban expansion, cropland drainage, and secondary effluent discharge indicates that nitrification of mixed agricultural and domestic nitrogen is the main mechanism. The transitional increase in sewage contribution from the upper to midstream zones mirrors the rapid peri-urban development along both river corridors, particularly in the YDR where urban sprawl and impervious surface expansion were more pronounced [36].

Cluster 3, comprising the downstream sites (YD2~YD7, CB1~CB2), represents the urban-sewage dominated regime. This cluster exhibited the highest NO_3^- and Cl^- concentrations, enriched $\delta^{15}N\text{-}NO_3^-$ values (14~19‰), and low DO, consistent with strong anthropogenic influence and partial denitrification. Sewage and manure contributed 61%, while fertilizer (17%) and soil N (15%) were secondary. The downstream YDR flows through densely built-up areas in Tianjin and Binhai, where untreated or partially treated wastewater and industrial effluents are likely dominant nitrate sources. In the CBR, the elevated sewage and manure input is associated with the inflow of the Yunchaojian River, which drains part of the Beijing urban area [64], as well as aquaculture activities near CB1 that release nitrogen from fish feed and organic residues [67]. These downstream segments thus represent hotspots of point-source pollution and in-stream nitrate transformation under low-oxygen conditions [86].

Across both rivers, nitrate sources showed a clear spatial shift from those associated with soil nitrogen and fertilizer in the upstream natural–agricultural zones to those dominated by sewage inputs in the downstream reaches. This pattern reflects increasing anthropogenic pressure and the gradual transition of land use along the Beijing–Tianjin–Hebei megacity gradient. The relatively stable yet persistent contribution of soil N (15~30%) across the clusters suggests that background mineralization continues to release nitrate even under strong human influence. Rainwater contributed less than 10% overall and remained a minor source in all clusters. The $\delta^{18}O\text{-}NO_3^-$ values (5–9 ‰) were far lower than those typical of atmospheric nitrate, confirming that precipitation played only a limited role compared with terrestrial inputs [87].

The SOM-MixSIAR integrated analysis refines the spatial identification of nitrate sources by combining unsupervised pattern recognition with probabilistic isotope modeling. The results highlight the coexistence of diffuse sources such as soil nitrogen and fertilizer and point sources such as sewage and manure, with the latter becoming dominant in downstream sections. These findings provide a scientific basis for targeted nitrogen management: improving urban wastewater treatment in the lower reaches, optimizing fertilizer use in agricultural midstream areas, and maintaining vegetation and buffer zones in upstream catchments to preserve background water quality.

To place the results in a broader context, comparisons with previous studies in China and internationally were conducted. Studies in Chinese megacity rivers (e.g., Haihe, Huangpu, and Pearl rivers) similarly report elevated nitrate concentrations and enriched $\delta^{15}N{-}NO_3^-$ values during the wet season [13], reflecting dominant anthropogenic inputs under monsoon-driven hydrological conditions. In contrast, urban rivers in Europe and North America generally exhibit lower nitrate levels and weaker isotopic enrichment [88–90], largely due to differences in hydrological regulation, seasonal sampling, and wastewater management. These comparisons indicate that while urbanization exerts a primary control on nitrate enrichment, local hydrological and environmental factors modulate nitrate sources and transformations. The combined hydrochemical–isotopic framework used here is therefore transferable when site-specific contexts are considered.

3.6. Nitrate Pollution Control in Megacity Rivers

As discussed above, nitrate pollution in the YDR and CBR shows a moderate level compared with many highly urbanized rivers in China [13], which may reflect the effectiveness of recent pollution-control actions in the Beijing–Tianjin–Hebei region [35,43]. Regional initiatives such as ecological water-replenishment projects and integrated watershed management have partially improved water quality [36,38]; however, the persistence of mixed nitrate sources highlights that current structural measures remain insufficient to achieve sustainable nitrogen control.

The dual-isotope and SOM-MixSIAR framework demonstrate that nitrate inputs are dominated by sewage and manure, particularly in the downstream urban reaches, while agricultural fertilizers and soil nitrogen remain important in the upper catchments. These patterns indicate that urban rivers in megacities are simultaneously affected by point-source discharges and diffuse non-point inputs, whose interaction is further complicated by hydrological regulation and seasonal flow variations [1]. Therefore, strategies focusing only on end-of-pipe treatment are unlikely to deliver long-term improvement.

For effective nitrate management in megacity rivers, a multi-scale approach combining source reduction, process enhancement, and ecological restoration is required [91]. In urban areas, improving wastewater collection and treatment capacity remains the most direct measure to reduce nitrogen loading, particularly through optimizing biological nutrient removal (BNR) processes such as anaerobic–anoxic–oxic continuous process (A^2/O), sequencing batch reactor (SBR), and membrane bioreactor (MBR) systems to enhance nitrification–denitrification efficiency [92,93]. In peri-urban and agricultural zones, optimizing fertilizer application and strengthening riparian vegetation can mitigate diffuse pollution [94]. Maintaining ecological base flow through reservoir regulation can also promote in-stream nitrification–denitrification coupling, enhancing the self-purification potential of the rivers [95].

Considering the complex human–natural interactions revealed in this study, future management of megacity rivers should emphasize integrated watershed governance, continuous isotopic monitoring, and adaptive regulation [96]. Such approaches would not only control nitrate pollution but also contribute to building a resilient and sustainable urban water environment.

3.7. Limitations and Future Perspectives

Despite the integrated hydrochemical and isotopic framework applied in this study, several limitations should be acknowledged. First, although the sampling sites covered the main stem of both river systems, the spatial density of sampling points was limited, particularly in some transitional reaches, which may not fully capture small-scale heterogeneity associated with localized inputs or in-stream processes. Second, water samples were collected during the wet season, when hydrological connectivity and nitrate transport are most active; therefore, seasonal variations in nitrate sources and transformation processes were not explicitly addressed. Third, nitrate source end-members were constrained using literature-reported isotopic signatures, and direct sampling of wastewater effluents was not conducted. While this approach is widely applied in regional-scale studies, direct characterization of local point sources would further reduce uncertainty in source apportionment. Future studies incorporating higher-density spatial sampling, multi-seasonal monitoring, and direct measurements of wastewater inputs would provide a more comprehensive understanding of nitrate dynamics in complex megacity river systems.

4. Conclusions

This study revealed the nitrate cycling mechanisms and pollution sources in the Yongding and Chaobai Rivers (YDR and CBR) within the Beijing–Tianjin–Hebei megacity region using hydrochemistry, dual nitrate isotopes ($\delta^{15}N$-NO_3^- and $\delta^{18}O$-NO_3^-), and a combined framework by Self-Organizing Map (SOM) and Bayesian mixing model (MixSIAR). Results showed clear spatial contrasts between the two rivers. In the YDR, nitrate and chloride concentrations increased downstream, indicating intensified anthropogenic influence, while the CBR displayed fluctuating patterns due to variable land use and reservoir regulation. Dual isotope data revealed that both point and non-point sources contributed to nitrate input, with widespread evidence of nitrification and spatially variable denitrification, particularly in low-oxygen, muddy-bottom reaches of the YDR. The NO_3^-/Cl^- ratio further supported the coexistence of urban wastewater and agricultural inputs. Results of SOM-MixSIAR framework revealed clear spatial variations in nitrate sources along the YDR and CBR. The SOM classified all sampling sites into 3 clusters, representing natural–agricultural, mixed, and sewage-dominated regimes, which together reflected a distinct transition from natural conditions upstream to increasing anthropogenic influence downstream. MixSIAR results showed that sewage and manure were the major nitrate sources, contributing 49.4% on average, followed by soil nitrogen (23.7%), chemical fertilizer (19.2%), and rainwater (7.7%). These findings suggest that nitrate pollution in megacity rivers is mainly driven by human activities, characterized by a downstream shift from agricultural and soil nitrogen inputs to sewage dominance, and jointly influenced by point- and non-point-source pollution as well as hydrological and land-use changes along the urbanization gradient. Effective management should combine wastewater upgrading, optimized fertilizer use, ecological flow maintenance, and watershed governance, supported by isotope-based monitoring and adaptive management to enhance nitrogen control and sustain urban aquatic ecosystems in the Beijing–Tianjin–Hebei megacity cluster.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/w18010106/s1, Table S1: Raw hydrochemical and nitrogen species data; Table S2: Nitrate isotope data.

Author Contributions: J.R.: Conceptualization, Investigation, Methodology, Writing—original draft, Writing—review, and editing; G.H.: Conceptualization, Project administration, Funding acquisition, Writing—review, and editing; X.L.: Writing—original draft, Writing—review, and editing; S.Z. and

X.G.: Investigation, Writing—review, and editing. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the "Deep-time Digital Earth" Science and Technology Leading Talents Team Funds for the Central Universities for the Frontiers Science Center for Deep-time Digital Earth, China University of Geosciences, Beijing (Fundamental Research Funds for the Central Universities; grant number: 2652023001).

Data Availability Statement: The original contributions presented in this study are included in the article and Supplementary Materials. Further inquiries can be directed to the corresponding author.

Acknowledgments: The authors gratefully acknowledge Jie Zeng from Guizhou University for the assistance with field sampling, Yunting Fang from Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, and Qian Zhang from the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, for the assistance with sample processing and laboratory analysis.

Conflicts of Interest: The authors declare no conflicts of interest.

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